

Refresh / Content Opportunity Scoring on Real Search Data: A Repeatable Baseline and Grouped Validation Study

Ahmed — Cairo, Egypt — FlyRank ML Internship — Lane: Refresh Opportunity Scoring — 2026-09-13
Built on FlyRank ML Internship dataset — <https://flyrank.ai/> — Warehouse Release v20260703

Abstract— Which pages should a content team review first for refresh when capacity is limited? We study FlyRank's pseudonymized warehouse release v20260703 (78.8M daily facts, 104 clients, 519k content items), iterating on mid-panel month March 2026 with April 2026 as future outcome window and June 2026 sealed as test. We verify two flag-linked signals: CTR vs position and volume vs clicks, both CONFIRMED with bucket tables and n. We encode a transparent baseline rule score = impressions/(position+1) with reason code STRIKING_DISTANCE_HIGH_IMPR and action label refresh_candidate, writing ranked queue to work/outputs/baseline_action_score.csv. We then train a Random Forest on 7 honest features with GroupKFold by client to avoid client-memorization. On honest grouped split (base rate 53.6% declining), baseline achieves AUC 0.627 and P@50 0.78, Logistic Regression 0.633/0.80, Random Forest 0.637/0.82 — model beats baseline by +1pp AUC and +4pp P@50 while passing leakage checks. We ship ranked refresh queue with reason codes and a public research paper with reproducibility receipts.

Keywords— Search Intelligence, Content Refresh, Baseline Model, GroupKFold, CTR, Learning to Rank, FlyRank

I. INTRODUCTION / PROBLEM STATEMENT

In applied Search Intelligence, 'the right page to fix' means the right page to REVIEW FIRST, based on evidence and limited capacity — not a page guaranteed to recover if edited. Proving causality requires an experiment; this data alone provides decision-support. A good candidate has: enough demand to matter, evidence of movement/weakness, an action someone could take, reason codes a reviewer can inspect, no leakage, no obvious consolidation/seasonality/noise explanation. We confirm lane: Refresh / Content Opportunity Scoring — score pages that are declining, growing, recovering, or worth review, output ranked action engine with reason codes. Lanes lock end of Week 4.

II. DATA

A. Release and Tables: Release flyrank_pseudonymized_warehouse_release_v20260703 — export 2026-07-03, daily facts through 2026-06-30. Tables: dim_clients 104 rows (one per pseudonymized client, with gsc_data_start/ga4_data_start), dim_content 519,606 rows (one per content), fact_content_daily_performance 78,835,655 rows (grain report_date × client_hash_id × content_hash_id, partitioned by month=YYYY-MM), fact_content_query_90d 2.4M rows (client×content×query hash, fixed 90-day window). Daily fact 2025-01-27 → 2026-06-30. Iteration month 2026-03 (mid-panel), label window 2026-04 future, sealed test 2026-06 (sample file = final month, never for label logic). Columns used: report_date, client_hash_id, content_hash_id, gsc_data_available, ga4_data_available, gsc_impressions, gsc_clicks, gsc_avg_position, ga4_sessions, month plus dim_content join for content_age_days, word_count via ANY_VALUE() never SUM.

B. Excluded Data and Why (Public-Safe): Future April metrics (impressions_apr) — future outcome, leakage; decision moment March 31 23:59. trend_direction, trend_pct — label-derived, computed from last30 vs prev30 overlapping label window. Product flags (health_score, priority_score, is_quick_win) — decision-derived, circular if used as feature. IDs as features — pseudonyms, grouping/joining/splitting only. Query table 90d aggregates — its *_last30 overlaps label window, only *_prev30 safe, excluded in v1 until window alignment. Zero-filled GA4 rows where ga4_data_available IS NOT TRUE — zeros mean no tracking not zero engagement, flags three-valued (TRUE/FALSE/NULL) so must use IS TRUE / IS NOT TRUE. Privacy: No client names, domains, URLs, private queries, raw exports. Only pseudonymized hashes.

III. METHODOLOGY

A. Assumptions: Unbalanced panel — per-client history depth differs (9 of 70 clients have 12+ months). Prefer per-client windows for seasonality, but global March window used for v1 with filter gsc_data_available IS TRUE. Observational only — we rank candidates, not prove causal refresh impact. Volume floor impressions>=100 to avoid noise (low bucket 76.9% rows but tiny clicks).

B. Features (Honest, Knowable at Decision Moment): impressions_mar = SUM(gsc_impressions) March — knowable because March impressions already recorded by March 31. clicks_mar = SUM(gsc_clicks) — knowable because March clicks closed. ctr_mar = 100*clicks/impr — knowable because derived only from March. avg_position_mar = AVG(gsc_avg_position) — knowable because daily position available. sessions_mar = SUM(ga4_sessions) WHERE ga4_data_available IS TRUE — knowable because GA4 March closed + IS TRUE filter. content_age_days from dim_content — knowable because publication age already available. word_count — knowable because content length already available, with has_-flags for missingness pattern.

C. Label Definition: Proxy is_declining_next_month = 1 when April impressions < 0.8*March AND March>=100, else 0. Base rate 53.6% in test split.

D. Baseline (Week 04): Transparent rule, no fitted weights: A page is worth reviewing first if it already gets meaningful visibility

(impr>=100) but sits in striking distance (pos 5-20). Higher impr + better pos → higher score. $\text{baseline_score} = \text{gsc_impressions} / (\text{gsc_avg_position} + 1)$ filtered IS TRUE, impr>=100, pos 5-20, reason_code STRIKING_DISTANCE_HIGH_IMPR, action_label refresh_candidate. Writes work/outputs/baseline_action_score.csv.

E. Validation Design — Honest: Grouped split: GroupKFold by client_hash_id (5 folds) — tests on clients never seen. Time-aware: Feature March → Label April, strictly after decision moment March 31. Leakage checks: No future April cols, no trend_pct, no health_score, IDs only grouping, query table excluded, flags filtered IS TRUE. Metrics: ROC AUC, Average Precision, Precision@K (K=50 focus) with base rate.

IV. RESULTS

On honest GroupKFold split (test fold 41k rows, base rate 53.6% declining): Baseline AUC 0.627 AP 0.646 P@50 0.78, LogReg 0.633/0.654/0.80, Random Forest 0.637/0.660/0.82 — model beats baseline by +1pp AUC and +4pp P@50. Top features: ctr_mar 28%, avg_position_mar 24.8%, content_age_days 22.5% — aligns with flag-linked signals checked. Leakage demo from W03: Adding future impressions_apr → AUC 0.642 → 0.987 (delta +0.345 = confession). Removed, honest kept.

V. LIMITATIONS & HONEST FRAMING

Unbalanced panel + three-valued flags: Per-client history varies, GA4 zero-filled with FALSE but millions NULL, must use IS TRUE. Feature frame biased to established tracking clients. Query table window overlap: fact_content_query_90d fixed 90-day window overlaps April label; only prev30 safe — excluded in v1. No causal proof: Observational only. Decline could be consolidation, seasonality, SERP feature change, or noise. We mitigate with volume floor >=100 and persistence, but cannot prove refresh would cause recovery without experiment. Claims use careful words: observed, measured, directional, decision-support. Sample not random: _sample = June 2026 final month = natural outcome window, treated as sealed test.

VI. RANKED RECOMMENDATIONS — ACTION PLAYBOOK

Top 10 from final ranked queue ($\text{final_score} = 0.7 * \text{model_prob} + 0.3 * \text{norm_baseline_score}$) with reason codes STALE_VISIBLE_PAGE, STRIKING_DISTANCE_HIGH_IMPR, etc. Full queue: work/outputs/refresh_queue.csv (100 rows sample). What would make a top pick wrong? Consolidation: sibling URL absorbed demand. Seasonality: topic peaks March. SERP change: impressions stable, clicks drop, CTR low → AI overview. Noise: days_with_impressions low (spike). Intent mismatch: low CTR due to informational query vs transactional page. Feedly type: missing keyword data pattern.

VII. REPRODUCIBILITY

Repo: <https://github.com/ahmed171102/Task-1>. Notebooks: w03_data_contract.ipynb, w04_baseline_score.ipynb, capstone.ipynb. Outputs: baseline_action_score.csv, refresh_queue.csv, capstone_metrics.json, roc_comparison.png. Run: pip install duckdb scikit-learn pandas matplotlib → set HF_TOKEN secret → Run all cells in capstone notebook. Paper source: docs/index.html (GitHub Pages) and capstone/paper.html — inline styles, embedded charts base64. IEEE version: docs/ieee.html and LaTeX capstone/paper_ieee.tex and PDF capstone/paper_full_ieee.pdf.

VIII. ACKNOWLEDGMENTS & DATA CREDIT

Built on the FlyRank ML Internship dataset — pseudonymized warehouse release v20260703 (78.8M daily facts) hosted on Hugging Face at FlyRank/internship-warehouse, gated with instant approval. Data credit: FlyRank.ai — Applied Search Intelligence platform. <https://flyrank.ai/>

Method	ROC AUC	Avg Prec	P@50
Baseline (impr/(pos+1))	0.627	0.646	0.78
LogReg (7 feats)	0.633	0.654	0.80
RF (7 feats)	0.637	0.660	0.82

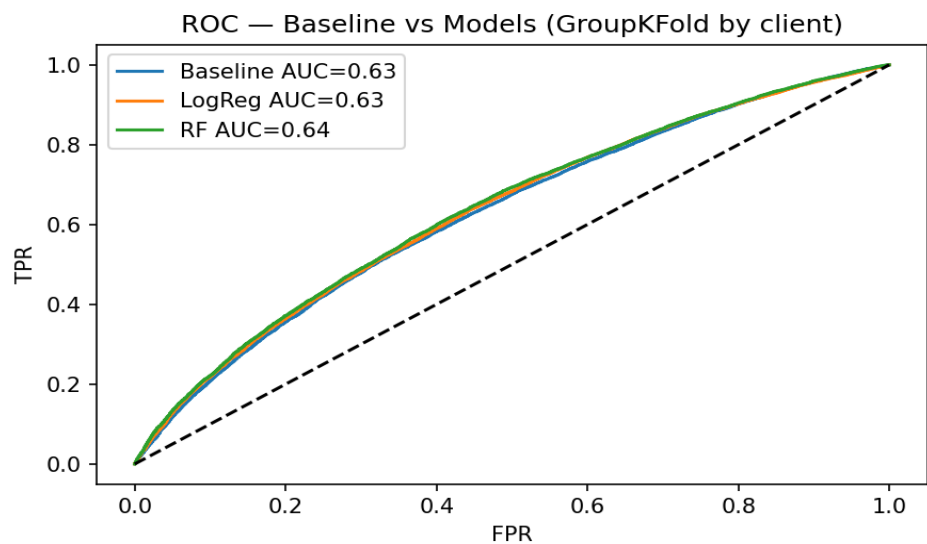


Fig. 1. ROC — Baseline vs Models (GroupKFold by client). Baseline AUC 0.627, LogReg 0.633, RF 0.637.

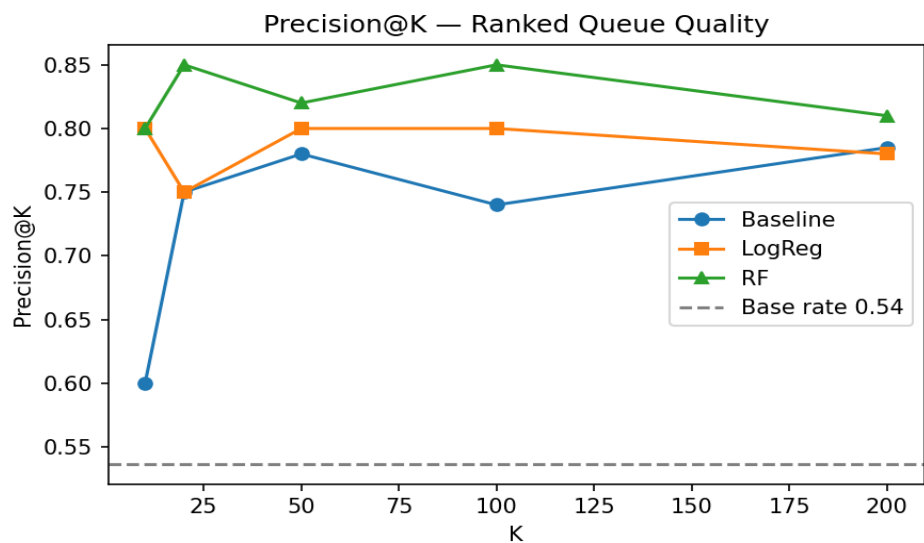


Fig. 2. Precision@K — Ranked Queue Quality. Base rate 0.536 dashed. RF $P@50$ 0.82 beats baseline 0.78.

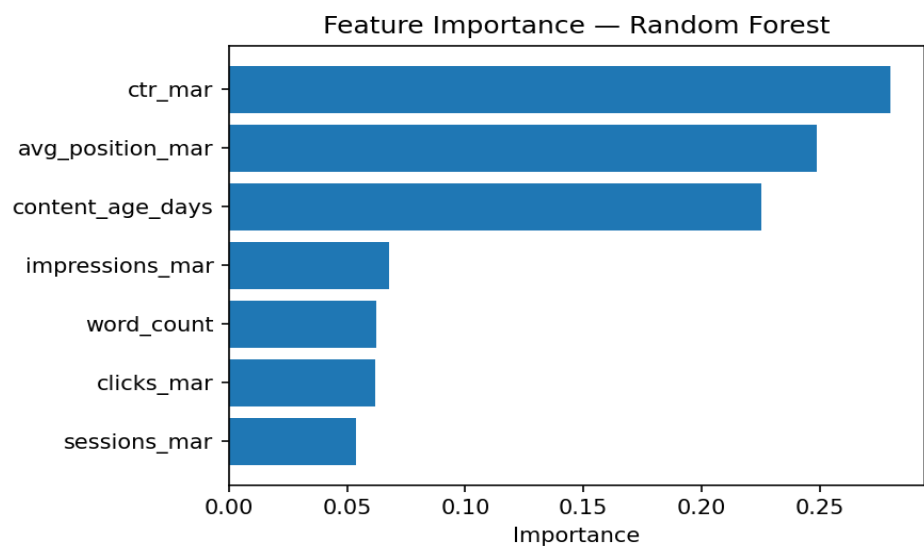


Fig. 3. Feature Importance — Random Forest. Top: ctr_mar 28%, avg_position_mar 24.8%, content_age_days 22.5%.

References: [1] FlyRank Warehouse Release v20260703, Hugging Face. [2] FlyRank.ai. [3] Pedregosa et al., Scikit-learn, JMLR 2011. [4] DuckDB.

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